

Analysis Report

1. Introduction

This analysis report presents the requirement analysis and system modeling phase of the Invisio graduation project. The objective of this document is to analyze the problem domain, examine existing solutions, and define the functional and non-functional requirements of the proposed system in a structured manner.

Invisio is designed as a web-based image editing system that combines AI-driven automation with limited manual refinement. Unlike systems that rely solely on either full automation or fully manual editing, Invisio adopts a hybrid human-in-the-loop approach in which artificial intelligence performs the main processing and users can correct minor inaccuracies when necessary.

This document serves as a foundation for the system design and implementation phases and ensures that the proposed solution is feasible, scalable, secure, and aligned with modern software engineering and AI system design principles.

2. Current system (if any)

Existing photo editing systems can be categorized into professional desktop applications and AI-based mobile or web applications.

Professional desktop tools provide fine-grained manual control but require advanced technical expertise, long learning periods, and time-consuming workflows. These tools are not accessible for non-technical users and are inefficient for quick editing tasks.

AI-based applications focus on full automation and ease of use. While these systems are fast and accessible, they rely entirely on AI predictions and do not allow users to correct errors. Inaccuracies in object boundaries, background detection, or segmentation results cannot be fixed without switching to professional tools.

Common limitations of current systems include insufficient control over AI outputs, poor handling of complex image regions such as hair or transparent objects, and privacy risks due to persistent storage of user images.

3. Proposed system

Invisio is proposed as a hybrid AI-assisted image editing system that integrates automated AI processing with lightweight manual refinement tools. The system first applies AI-based models for tasks such as object detection, segmentation, inpainting, and enhancement.

If minor inaccuracies occur, users can refine AI-generated masks using brush-based tools. Manual interaction is restricted to mask-level corrections rather than direct pixel-level editing, ensuring simplicity and performance.

The system is implemented as a web-based platform and does not require installation. All image data is processed temporarily during the session and removed immediately after completion.

3.1 Overview

The system follows a modular architecture consisting of a frontend interface, a backend processing layer, an AI inference layer, and a session management component.

The frontend enables users to upload images, select editing features, refine results, preview outputs, and download final images. The backend routes requests and manages processing workflows. The AI layer performs segmentation and enhancement operations. Session management ensures temporary handling of image data without persistent storage.

This structure improves maintainability, scalability, and extensibility.

3.2 Functional Requirements

The system shall support the following functional requirements:

- FR1 – The system shall allow users to upload image files via a web interface.
- FR2 – The system shall automatically detect and remove unwanted objects from images.
- FR3 – The system shall enhance brightness, contrast, and color using AI-based models.
- FR4 – The system shall apply artistic styles through neural style transfer.
- FR5 – The system shall beautify facial features in portrait images.
- FR6 – The system shall replace or blur image backgrounds automatically.
- FR7 – The system shall colorize black-and-white images using AI.
- FR8 – The system shall display a preview of processed images.
- FR9 – The system shall allow users to download edited images.

- FR10 – The system shall process image data temporarily and delete it after completion.
- FR11 – The system shall allow manual refinement of AI-generated masks using brush-based tools.
- FR12 – The system shall support a hybrid editing workflow combining AI automation with manual refinement.

3.3 Nonfunctional Requirements

- Performance: AI inference operations shall complete within acceptable response times. Manual refinement tools shall operate only on AI-generated masks to minimize computational overhead.
- Security: All communication shall be implemented using HTTPS with TLS 1.3. Backend APIs shall follow OWASP API Security Top 10 guidelines, including request validation and token-based access control.
- Data Privacy: Images shall be processed exclusively in volatile memory and discarded immediately after request completion. The system design aligns with GDPR Article 25 and KVKK Article 12 by enforcing data minimization and privacy-by-design principles.
- Usability: The interface shall be intuitive for non-technical users. Manual tools shall appear only when necessary and shall not expose professional-level editing complexity.
- Portability: The system shall function across modern web browsers without requiring installation.

3.4 Pseudo requirements

The following pseudo requirements represent qualitative expectations that are difficult to measure precisely but essential for system success:

- The system should feel responsive during interaction.
- AI-generated edits should appear visually realistic.
- Users should feel in control without being overwhelmed.
- Manual refinement should feel corrective rather than creative.

3.5 System models

The system behavior is modeled using standard software engineering representations.

System modeling approach includes:

- Use case models to describe user-system interactions
- Scenario descriptions for typical user workflows
- Conceptual object and class models
- Dynamic models illustrating interaction flow

These models help ensure a clear understanding of system behavior before implementation.

3.5.1 Scenarios

Scenario 1: In this scenario, the user interacts with the system using a fully automated workflow. The user uploads an image through the web interface and selects one of the available AI-based editing features, such as background removal or image enhancement. The system validates the uploaded image, routes it to the appropriate AI processing module, and generates the edited result automatically. Once processing is completed, the system displays a preview of the edited image. The user reviews the result and proceeds to download the final output without performing any manual adjustments.

This scenario represents the default usage pattern for users who are satisfied with AI-generated results and do not require additional control.

Scenario 2: In this scenario, the user follows a hybrid editing workflow that combines AI automation with manual refinement. The user uploads an image and selects an AI-based feature. After the system completes the initial AI processing, the user notices minor inaccuracies in the output, such as imperfect object boundaries or segmentation errors.

The system allows the user to refine the AI-generated mask using a brush-based tool. These refinements are applied only to the mask layer and do not involve direct pixel-level editing. Once the refinement is completed, the system reprocesses the image using the corrected mask and generates an improved final result. The user previews the updated output and downloads the image.

This scenario demonstrates the system's human-in-the-loop design, enabling higher accuracy while maintaining usability.

Scenario 3: In this scenario, the user chooses to share an edited image with the community after completing the editing process. Once the final image is generated and previewed, the system presents an optional sharing action. The user explicitly selects the option to share the image.

Before publishing, the system requests user confirmation and removes any session-related or personal metadata from the image. The shared image is then

published to a public exploration section where other users can browse, view, and gain inspiration from community-shared content.

This scenario is entirely optional and does not affect the core editing workflow. Users who do not opt in to sharing can complete all editing tasks without interacting with the community features. Shared images are treated as public content and are not used for model training unless explicitly authorized.

This scenario supports discoverability and user engagement while preserving user control and privacy.

Scenario 4: In this scenario, the user uploads an image that contains content violating the system's content safety policies. During the upload phase, the system automatically performs an NSFW content check as part of mandatory validation procedures.

If the content is classified as inappropriate, the system halts further processing and informs the user that the image cannot be edited due to policy restrictions. The image is not forwarded to any AI processing modules and is immediately discarded from volatile memory.

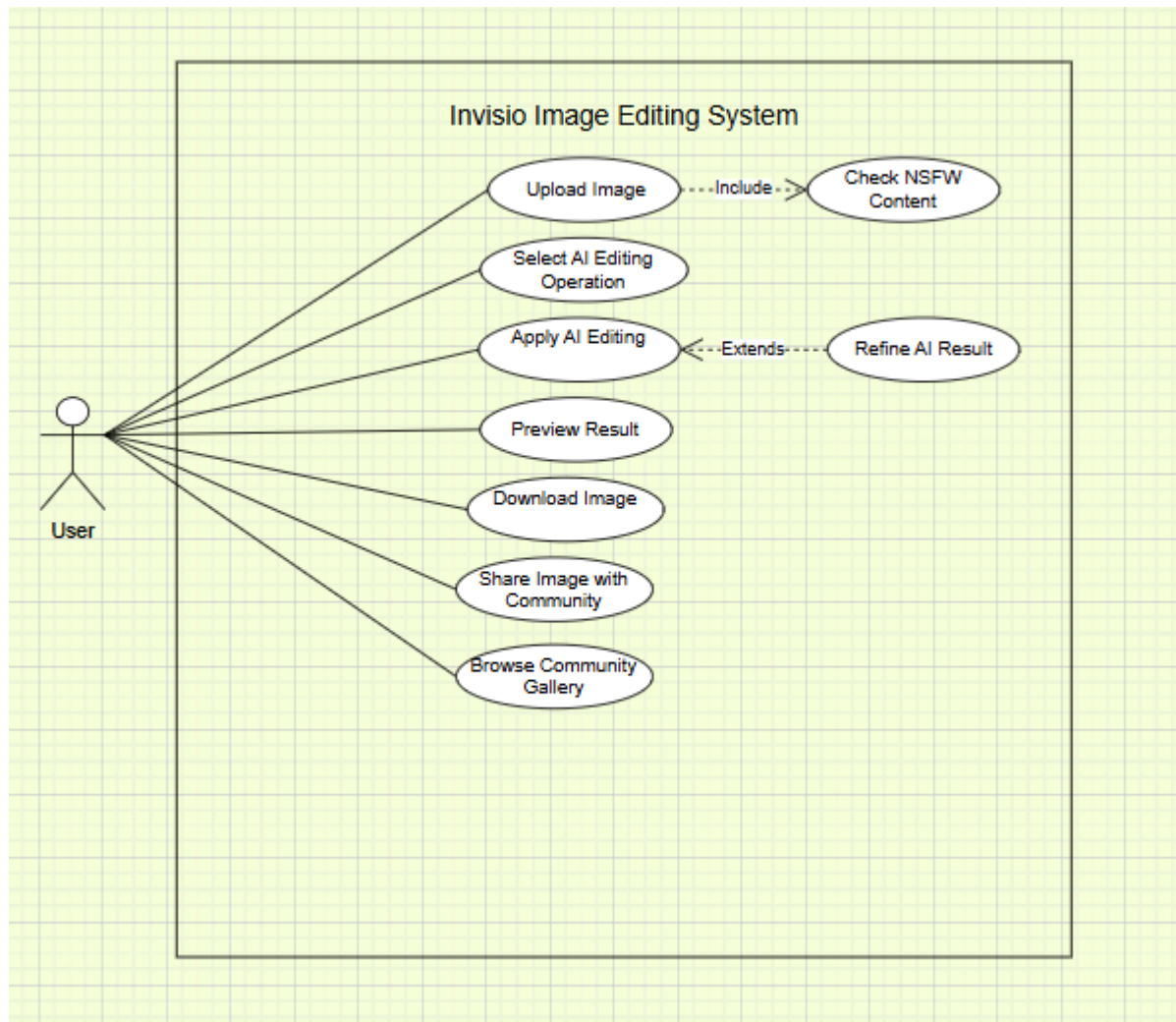
This scenario highlights the system's ethical safeguards and responsible content handling mechanisms, ensuring compliance with platform policies and legal considerations.

Scenario 5: In this scenario, the user uploads an image that does not meet the system's input requirements, such as an unsupported file format or exceeding the maximum file size limit. Upon receiving the upload request, the system performs input validation checks before initiating any AI-based processing.

The system detects the invalid input and rejects the request without forwarding it to the AI processing layer. A clear and informative error message is displayed to the user, indicating the reason for the rejection and guiding the user toward corrective action. No image data is processed or retained beyond the validation stage.

This scenario demonstrates the system's ability to handle invalid inputs safely and efficiently, preventing unnecessary resource consumption and ensuring predictable system behavior.

3.5.2 Use case model



3.5.3 Object and class model

The main conceptual objects in the system include **User**, **Image**, **EditingRequest**, **AIModel**, **Mask**, **RefinementTool**, **ResultImage**, and **Session**.

- **User**: Represents the end user interacting with the system.

- **Image:** Represents the uploaded input image.
- **EditingRequest:** Encapsulates the selected editing operation and associated parameters.
- **AIModel:** Represents the AI model used for processing tasks such as segmentation or enhancement.
- **Mask:** Represents the AI-generated or user-refined region map used during processing.
- **RefinementTool:** Represents the brush-based tool used for mask correction.
- **ResultImage:** Represents the final processed image.
- **Session:** Manages temporary data and controls the lifecycle of image processing.

This object model provides a conceptual foundation for system implementation and data flow.

3.5.4 Dynamic models

Dynamic behavior of the system includes:

- Image upload and validation
- Routing to AI processing modules
- Result generation and preview
- Cleanup of temporary data

These interactions ensure a controlled and secure execution flow.

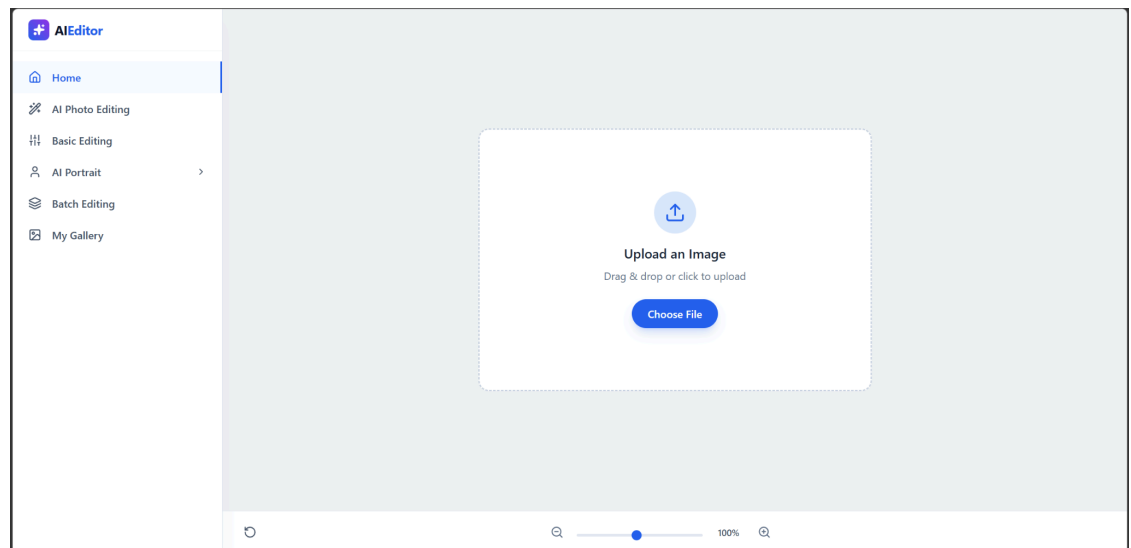
3.5.5 User interface - navigational paths and screen mock-ups

Navigation flow:

- Home Page → Image Upload
- Editing Options Selection
- Preview Screen
- Download Screen

The UI is designed to minimize navigation steps and maintain clarity.

The image below contains a prototype/demo of UI. The demo is setted for usage as a kind of template. Further implementations and innovations will be applied to that user interface template.



Screenshot displays home screen and image upload container. Sidebar contains and shows editing options and selections. Editing section divided for splitting ai

assisted editing and basic(filter applications etc.) editing. Images will be stored at my gallery section. Further features will be added with sprints.

4. Glossary

- **AI (Artificial Intelligence):**
Computational techniques that enable machines to perform tasks requiring human intelligence, such as image understanding and enhancement.
- **CNN (Convolutional Neural Network):**
A deep learning architecture commonly used for visual feature extraction and image analysis.
- **GAN (Generative Adversarial Network):**
A generative model composed of competing neural networks used to synthesize realistic images.
- **Segmentation:**
The process of partitioning an image into meaningful regions at the pixel level.
- **Inpainting:**
A technique for reconstructing removed or missing image regions using surrounding context.
- **Mask:**
A region map used to guide AI processing and manual refinement.
- **Human-in-the-loop:**
A system design approach that incorporates limited human input to improve automated AI outputs.
- **Hybrid Editing Workflow:**
An editing pipeline combining AI automation with optional manual refinement.
- **TLS (Transport Layer Security):**
A cryptographic protocol used to secure data transmission.

- **GDPR (General Data Protection Regulation):**
A European Union regulation governing personal data protection.
- **KVKK (Kişisel Verilerin Korunması Kanunu):**
Türkiye's law regulating the protection of personal data.

5. References

- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., et al. (2014). *Generative adversarial nets*. NeurIPS.
- Gatys, L. A., Ecker, A. S., & Bethge, M. (2015). *A neural algorithm of artistic style*. arXiv preprint.
- Jing, Y., Yang, Y., Feng, Z., Ye, J., Yu, Y., & Song, M. (2017). *Neural style transfer: A review*. IEEE Trans. on Visualization and Computer Graphics.
- Xu, Y. (2025). *Style transfer review: Traditional machine learning to deep learning*. Information, 16(2).
- Zhang, S., et al. (2025). *Applying deep learning for style transfer in digital art*. Scientific Reports.
- Diamant, N., Zadok, D., Baskin, C., Schwartz, E., & Bronstein, A. M. (2019). *Beholder-GAN: Generation and beautification of facial images with conditioning on their beauty level*. In *Proc. IEEE ICIP*.
- Gan, J., et al. (2024). *Applied research on face image beautification based on FIBGAN*. Electronics.
- Peng, T. (2024). *ISFB-GAN: Interpretable semantic face beautification*. Expert Systems with Applications.
- Ronneberger, O., Fischer, P., & Brox, T. (2015). *U-Net: Convolutional networks for biomedical image segmentation*. MICCAI.
- Ling, H., Kreis, K., Li, D., Kim, S. W., Torralba, A., & Fidler, S. (2021). *EditGAN: High-precision semantic image editing*. arXiv preprint.
- Anwar, S., Tahir, M., Li, C., Mian, A., Khan, F. S., & Muzaffar, A. W. (2020). *Image colorization: A survey and dataset*. arXiv preprint.
- [Author(s)] (2024). *[Survey on image colorization]*. Electronics.
- Gui, J., Sun, Z., Wen, Y., Tao, D., & Ye, J. (2020). *A review on Generative Adversarial Networks: Algorithms, theory, and applications*.

- Shamsolmoali, P., Zareapoor, M., Granger, E., Zhou, H., Wang, R., Celebi, M. E., & Yang, J. (2020). *Image synthesis with adversarial networks: A comprehensive survey and case studies*. arXiv preprint.
- Wu, X., Xu, K., & Hall, P. M. (2017). *A survey of image synthesis and editing with generative adversarial networks*. Tsinghua Sci. & Tech.
- Zhou, X. (2025). *Survey paper bridging the metrics gap in image style transfer*. [Journal].
- Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., & Hochreiter, S. (2017). *GANs trained by a two time-scale update rule converge to a local Nash equilibrium*. NeurIPS.
- [Author(s)] (2025). *A review on generative AI for text-to-image and image-to-image generation*. arXiv preprint.